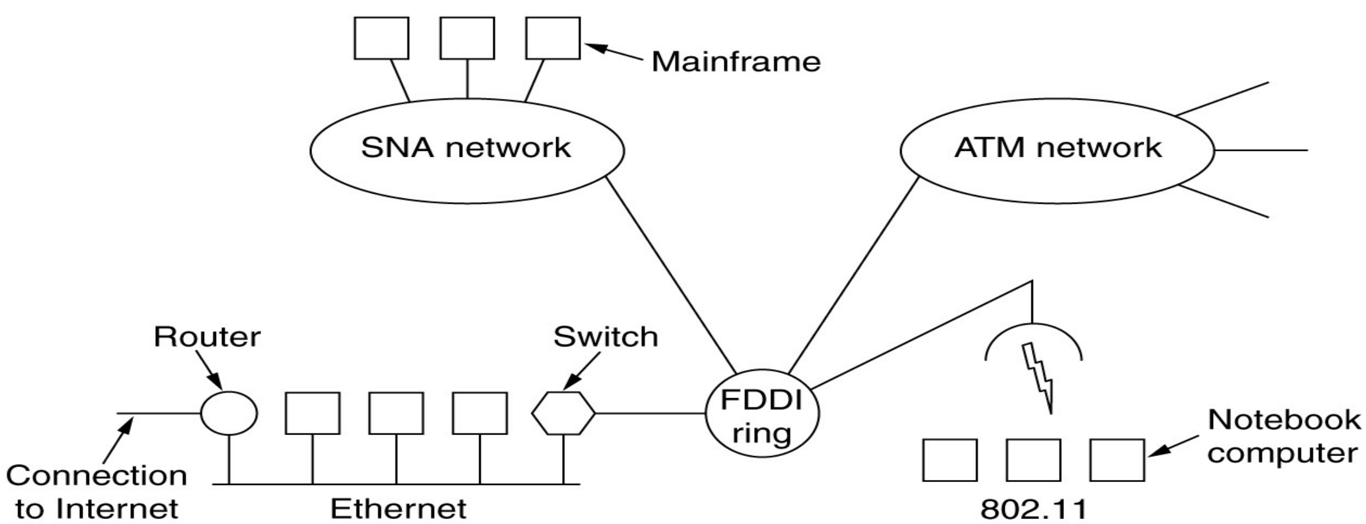
UNIT IV

INTERNETWORKING

- **Internetworking** started as a way to connect types of computer networking technology. Computer network term is used to describe two or more computers that are linked to each other. When two or more computer networks or computer network segments are connected using devices such as a router then it is called as **computer internetworking**.

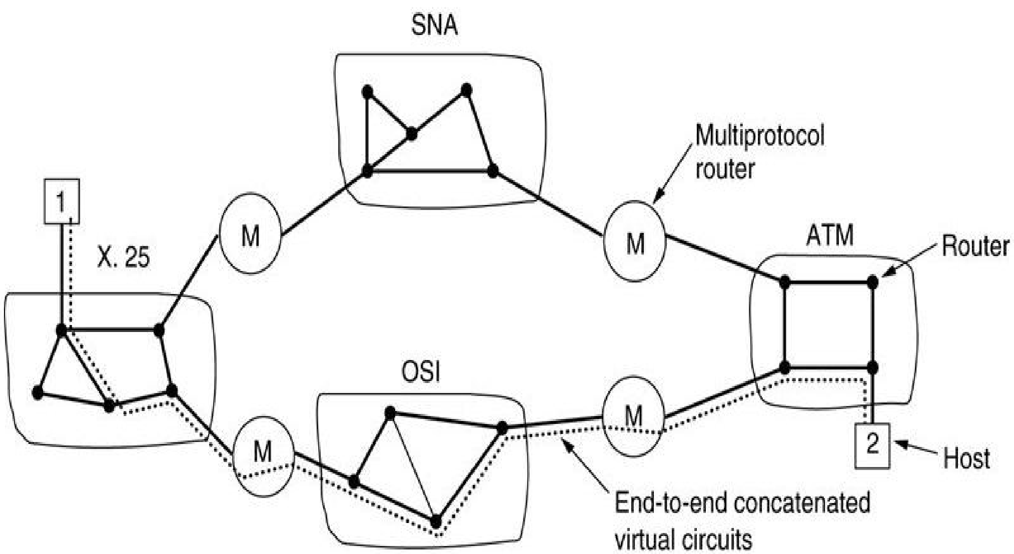
  A single homogeneous network, with each machine using the same protocol in each layer. Unfortunately,
- this assumption is wildly optimistic. Many different networks exist, including LANs, MANs, and WANs.
- Numerous protocols are in widespread use in every layer. In the following sections we will take a careful
- look at the issues that arise when two or more networks are connected to form an internet.

- As an example of how different networks might be connected, consider the example of Fig Here we see a corporate network with multiple locations tied together by a wide area ATM network. At one of the locations, an FDDI optical backbone is used to connect an Ethernet, an 802.11 wireless LAN, and the corporate data center's SNA mainframe network.
- The purpose of interconnecting all these networks is to allow users on any of them to communicate with users on all the other ones and also to allow users on any of them to access data on any of them. Accomplishing this goal means sending packets from one network to another.



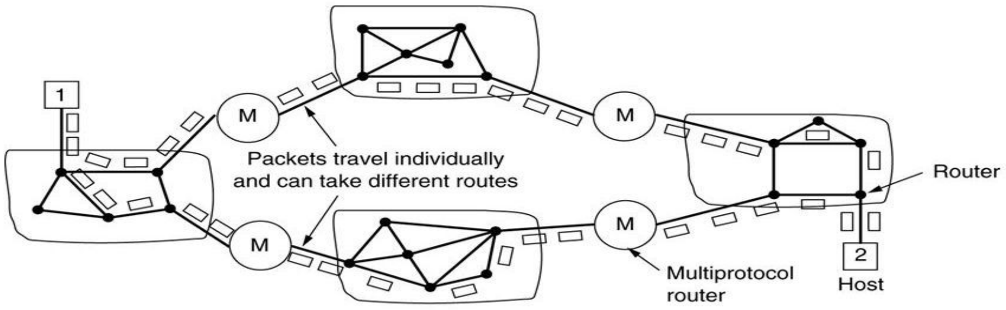
- **Systems Network Architecture (SNA) Network**
- A data communication architecture created by IBM in the 1970s to standardize communication among its mainframes and other hardware.
- It divides network functions into a hierarchical structure of layers to enable devices to communicate effectively without detailed knowledge of each other.
- An ATM (**Asynchronous Transfer Mode**) network is a high-speed, connection-oriented networking technology that uses fixed-size cells to transmit data, voice, and video. It was designed to provide predictable data delivery, reduce delays, and ensure quality of service (QoS) by breaking data into 53-byte cells (5 bytes header, 48 bytes data) and routing them through virtual circuits.
- **Fiber Distributed Data Interface**. It is a set of ANSI and ISO guidelines for information transmission on fiber-optic lines in Local Area Network (LAN) that can expand in run upto 200 km (124 miles). The FDDI convention is based on the **token ring protocol.**
- In expansion to being expansive geographically, an FDDI neighborhood region arranges can support thousands of clients. FDDI is habitually utilized on the spine for a Wide Area Network(WAN).

- **Concatenated Virtual Circuits**
- Two styles of internetworking are possible: a connection-oriented concatenation of virtual circuit subnets, and
- a datagram internet style. We will now examine these in turn, but first a word of caution. In the past,
- most (public) networks were connection oriented (and frame relay, SNA, 802.16, and ATM still are).



- X.25 is a connection-oriented, packet-switching protocol standard for wide-area networks (WANs) that was developed by the International Telecommunication Union-T (ITU-T). It established virtual circuits to ensure reliable, ordered data delivery between user devices and network equipment .
- In the concatenated virtual-circuit model, shown in Fig a connection to a host in a distant network is set up in a way similar to the way connections are normally established. The subnet sees that the destination is remote and builds a virtual circuit to the router nearest the destination network. Then it constructs a virtual circuit from that router to an external gateway (multiprotocol router).
- This gateway records the existence of the virtual circuit in its tables and proceeds to build another virtual circuit to a router in the next subnet. This process continues until the destination host has been reached.

- **Connectionless Internetworking**
- The alternative internetwork model is the datagram model, shown in Fig In this model, the only service the network layer offers to the transport layer is the ability to inject datagrams into the subnet and hope for the best. There is no notion of a virtual circuit at all in the network layer, let alone a concatenation of them. This model does not require all packets belonging to one connection to traverse the same sequence of gateways.

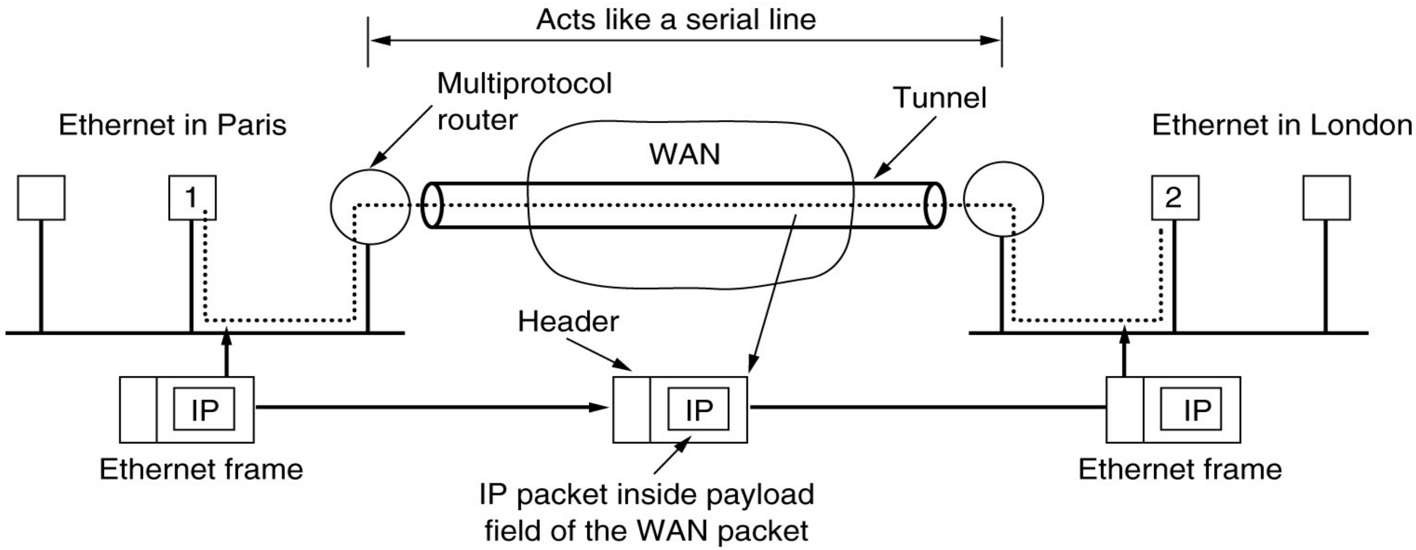


A connectionless internet.

- In Fig datagrams from host 1 to host 2 are shown taking different routes through the internetwork. A routing decision is made separately for each packet, possibly depending on the traffic at the moment the packet is sent. This strategy can use multiple routes and thus achieve a higher bandwidth than the concatenated virtual-circuit model.

- **Internet**
- Scope: A vast, public, global network of computers.
- Users: Accessible to anyone with the proper connection.
- Information: Public and general information, though also used for e-commerce.
- Access: Unlimited public access with no restrictions.
- **Intranet**
- Scope: A private, internal network within a single organization.
- Users: Only authorized employees of the organization.
- Information: Specific, corporate, and proprietary information for internal use.
- Access: Private and restricted to internal users, often requiring a password and a secure connection like a VPN.
- **Extranet**
- Scope: An extended version of an organization's intranet, linking to specific authorized users outside the company.
- Users: Authorized external partners, such as vendors, clients, or suppliers.
- Information: Shared information and files with specific authorized collaborating groups.
- Access: Private and controlled, allowing external partners to access selected information without full access to the entire network.

- **Tunneling**
- A technique of inter-networking called **Tunneling** is used when source and destination networks of the same type are to be connected through a network of different types. Tunneling uses a layered protocol model such as those of the OSI or TCP/IP protocol suite.
- So, in other words, when data moves from host A to B it covers all the different levels of the specified protocol (OSI, TCP/IP, etc.) while moving between different levels, data conversion (Encapsulation) to suit different interfaces of the particular layer is called tunneling.
- **Steps**
- Host A constructs a packet that contains the IP address of Host B.
- It then inserts this IP packet into an Ethernet frame and this frame is addressed to the multiprotocol router M1
- Host A then puts this frame on Ethernet.
- When M1 receives this frame, it removes the IP packet, inserts it in the payload packet of the WAN network layer packet, and addresses the WAN packet to M2. The multiprotocol router M2 removes the IP packet and sends it to host B in an Ethernet frame.



- For example, let us consider an Ethernet to be connected to another Ethernet through a WAN as:
- **How Does Encapsulation Work?**
- Data travels from one place to another in the form of packets, and a packet has two parts, the first one is the header which consists of the destination address and the working protocol and the second thing is its contents.
- In simple terminology, Encapsulation is the process of adding a new packet within the existing packet or a packet inside a packet. In an encapsulated packet, the header part of the first packet is remain surrounded by the payload section of the surrounding packet, which has actual contents.

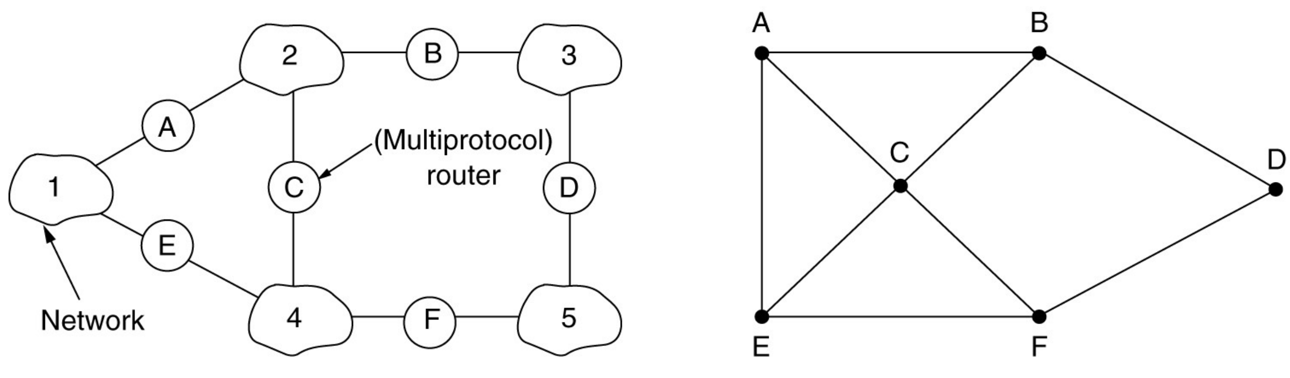
- **Generic Routing Encapsulation (GRE):**Generic Routing Encapsulation is a method of encapsulation of IP packets in a GRE header that hides the original IP packet. Also, a new header named delivery header is added above the GRE header which contains the new source and destination address.
- GRE header act as a new IP header with a Delivery header containing a new source and destination address. Only routers between which GRE is configured can decrypt and encrypt the GRE header. The original IP packet enters a router, travels in encrypted form, and emerges out of another GRE-configured router as the original IP packet as they have traveled through a tunnel. Hence, this process is called GRE tunneling.
- **Internet Protocol Security (IPsec):**IP security (IPSec) is an Internet Engineering Task Force (IETF) standard suite of protocols between 2 communication points across the IP network that provide data authentication, integrity, and confidentiality. It also defines the encrypted, decrypted, and authenticated packets. The protocols needed for secure key exchange and key management are defined in it.

IP-in-IPIP-in-IP is a Tunneling Protocol for encapsulating IP packets inside another IP packet.

- **Secure Shell (SSH):**SSH(Secure Shell) is an access credential that is used in the SSH Protocol. In other words, it is a cryptographic network protocol that is used for transferring encrypted data over the network. It allows you to connect to a server, or multiple servers, without having to remember or enter your password for each system which is to log in remotely from one system to another.

Point-to-Point Tunneling Protocol (PPTP):PPTP or Point-to-Point Tunneling Protocol generates a tunnel and confines the data packet. Point-to-Point Protocol (PPP) is used to encrypt the data between the connection. PPTP is one of the most widely used VPN protocols and has been in use since the early release of Windows.

- **Internetwork Routing**
- Internetworking is the process or technique of connecting different networks by using intermediary devices such as routers or gateway devices. Internetworking ensures data communication among networks owned and operated by different entities using a common data communication and the Internet Routing Protocol.

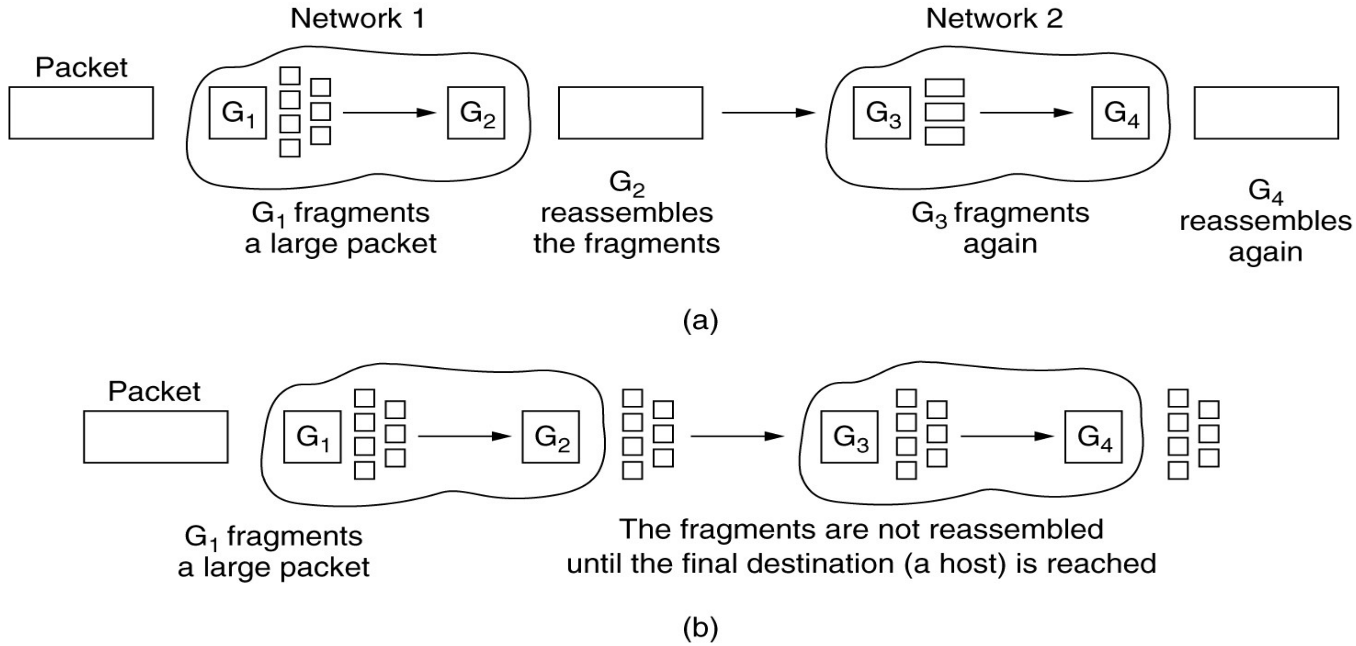


(a) (b)

:twork

- Every node or segment of a network is constructed using a similar protocol or communication logic such as Transfer Control Protocol (TCP) or Internet Protocol (IP), to enable communication.
- When one network interacts with another network using ongoing communication protocols, it is termed as 'internetworking'. Internetworking was designed to address the challenge of delivering a packet of data across multiple links.
- The difference between network expansion and internetworking is subtle. Using a switch or hub to connect two local area networks is a simple extension of a LAN, whereas connecting them via a router is an example of internetworking.
- Internetworking is enforced by the Network Layer (Layer 3) of the OSI-ISO model. The most well-known example of internetworking is the internet.

- **Fragmentation**
- Most Ethernet segments have their maximum transmission unit (MTU) fixed to 1500 bytes. A data packet can have more or less packet length depending upon the application.
- Devices in the transit path also have their hardware and software capabilities which tell what amount of data that device can handle and what size of packet it can process.
- If the data packet size is less than or equal to the size of packet the transit network can handle, it is processed neutrally. If the packet is larger, it is broken into smaller pieces and then forwarded. This is called packet fragmentation.
- Each fragment contains the same destination and source address and routed through transit path easily. At the receiving end it is assembled again.



(a) Transparent fragmentation. (b) Nontransparent fragmentation.

Transparent Fragmentation - the router doing the fragmentation breaks up oversized packets and addresses them all to the same exit router. The exit router re-assembles the fragments and forwards original packet. This is called **transparent fragmentation because it is invisible to the other networks in the path and to the hosts**.

Problems in Transparent Fragmentation:

1) All packets must exist the same router; doesn't work for a connectionless network.

2) Overhead and delay involved with repeated fragmentation and reassembly.

- **Nontransparent Fragmentation** - once a packet is fragmented, it is not reassembled until it reaches the destination.

  Problems in Nontransparent Fragmentation:

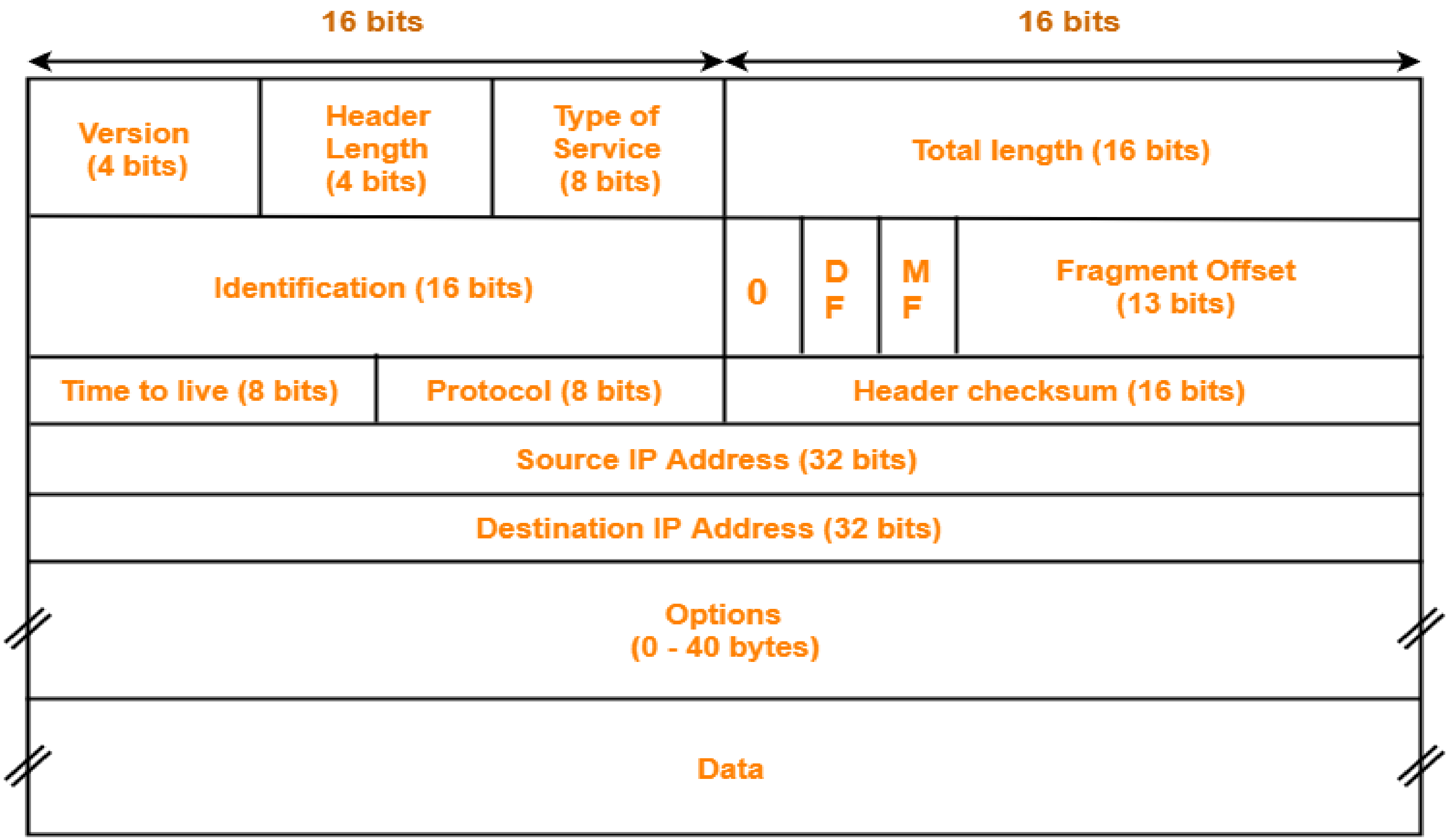
  1) Overhead is increased since the ratio of data to header is worse.

  2) Every host must be able to reassemble.

  3)Multiple layers of fragmentation are possible. (Some networks may need to fragment the fragments.)

- **Internet Protocol Version 4 (IPv4)**
- **The IPV4 Protocol**
- An appropriate place to start our study of the network layer in the Internet is the format of the IP datagrams themselves. An IP datagram consists of a header part and a text part. The header has a 20-byte fixed part and a variable length optional part.
- The header format is shown in Fig. 5-53. It is transmitted in big-endian order: from left to right, with the high-order bit of the Version field going first. (The SPARC is big endian; the Pentium is little-endian.) On little endian machines, software conversion is required on both transmission and reception.

- # IPV4 HEADER



IPv4 Header

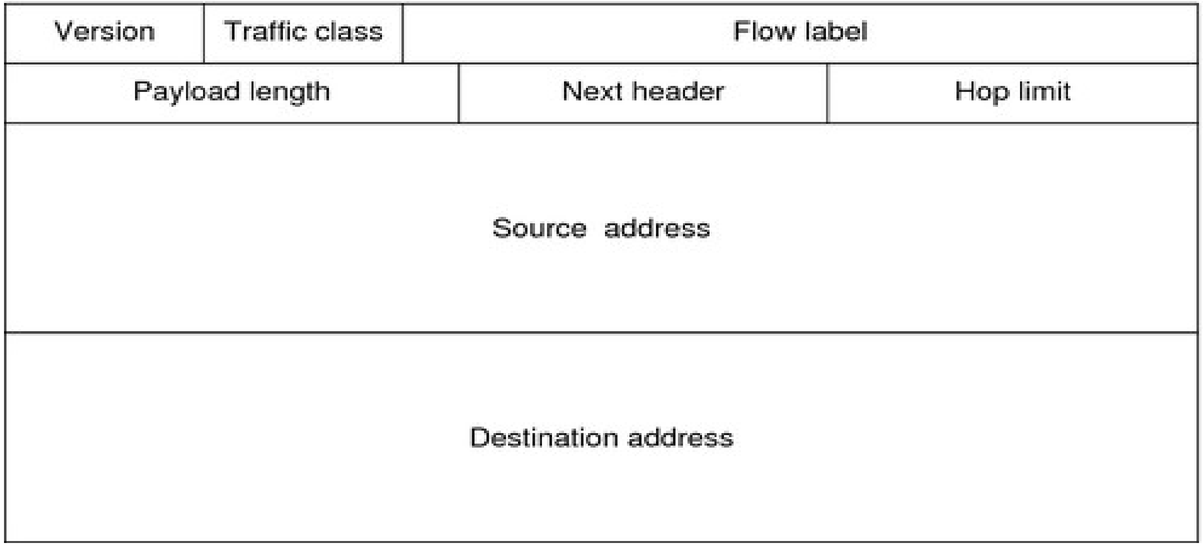
- **Version field** :The Version field keeps track of which version of the protocol the datagram belongs to. By including the version in each datagram, it becomes possible to have the transition between versions take years, with some machines running the old version and others running the new one. Currently a transition between IPv4 and IPv6 is going on, has already taken years, and is by no means close to being finished.
- **Header length** :Since the header length is not constant, a field in the header, IHL, is provided to tell how long the header is, in 32-bit words. The minimum value is 5, which applies when no options are present. The maximum value of this 4-bit field is 15, which limits the header to 60 bytes, and thus the Options field to 40 bytes.
- **Type of service** :The Type of service field is one of the few fields that has changed its meaning (slightly) over the years. It was and is still intended to distinguish between different classes of service. Various combinations of reliability and speed are possible. For digitized voice, fast delivery beats accurate delivery. For file transfer, error-free transmission is more important than fast transmission.
- Originally, the 6-bit field contained (from left to right), a three-bit Precedence field and three flags, D, T, and R. The Precedence field was a priority, from 0 (normal) to 7 (network control packet). The three flag bits allowed the host to specify what it cared most about from the set {Delay, Throughput, Reliability}.

- **Total length** :The Total length includes everything in the datagram—both header and data. The maximum length is 65,535 bytes. At present, this upper limit is tolerable, but with future gigabit networks, larger datagrams may be needed.
- **Identification field** :The Identification field is needed to allow the destination host to determine which datagram a newly arrived fragment belongs to. All the fragments of a datagram contain the same Identification value Next comes an unused bit and then two 1-bit fields.
- **DF** stands for Don't Fragment. It is an order to the routers not to fragment the datagram because the destination is incapable of putting the pieces back together again.
- **MF** stands for More Fragments. All fragments except the last one have this bit set. It is needed to know when all fragments of a datagram have arrived.
- **Fragment offset** :The Fragment offset tells where in the current datagram this fragment belongs. All fragments except the last one in a datagram must be a multiple of 8 bytes, the elementary fragment unit. Since 13 bits are provided, there is a maximum of 8192 fragments per datagram, giving a maximum datagram length of 65,536 bytes, one more than the Total length field.
- **Time to live field** :The Time to live field is a counter used to limit packet lifetimes. It is supposed to count time in seconds, allowing a maximum lifetime of 255 sec. It must be decremented on each hop and is supposed to be decremented multiple times when queued for a long time in a router. In practice, it just counts hops. When it hits zero, the packet is discarded and a warning packet is sent back to the source host.

- **Protocol field** :The Protocol field tells it which transport process to give it to. TCP is one possibility, but so are UDP and some others. The numbering of protocols is global across the entire Internet.
- **Header checksum** :The Header checksum verifies the header only. Such a checksum is useful for detecting errors generated by bad memory words inside a router. The algorithm is to add up all the 16-bit halfwords as they arrive, using one's complement arithmetic and then take the one's complement of the result. For purposes of this algorithm, the Header checksum is assumed to be zero upon arrival.
- **Source address and Destination address**: The Source address and Destination address indicate the network number and host number. We will discuss Internet addresses in the next section. The Options field was designed to provide an escape to allow subsequent versions of the protocol to include information not present in the original design, to permit experimenters to try out new ideas, and to avoid allocating header bits to information that is rarely needed.
- **Options**: The options are variable length. Each begins with a 1-byte code identifying the option. Some options are followed by a 1-byte option length field, and then one or more data bytes.

- **IPV6**
- **IPv6 Packet Header Format**
- The IPv6 protocol defines a set of headers, including the basic IPv6 header and the IPv6 extension headers.
- The following figure shows the fields that appear in the IPv6 header and the order in which the fields appear.

  - **Figure IPv6 Basic Header Format**



1.Version (4-bits): It represents the version of Internet Protocol, i.e. 0110.

2. **Traffic Class** (8-bits): These 8 bits are divided into two parts. The most significant 6 bits are used for Type of Service to let the Router Known what services should be provided to this packet. The least significant 2 bits are used for Explicit Congestion Notification (ECN).

3.Flow Label (20-bits): This label is used to maintain the sequential flow of the packets belonging to a communication. The source labels the sequence to help the router identify that a particular packet belongs to a specific flow of information. This field helps avoid re-ordering of data packets. It is designed for streaming/real-time media.

4. **Payload Length** (16-bits): This field is used to tell the routers how much information a particular packet contains in its payload. Payload is composed of Extension Headers and Upper Layer data. With 16 bits, up to 65535 bytes can be indicated; but if the Extension Headers contain Hop-by-Hop Extension Header, then the payload may exceed 65535 bytes and this field is set to 0.

5. **Next Header** (8-bits): This field is used to indicate either the type of Extension Header, or if the Extension Header is not present then it indicates the Upper Layer PDU. The values for the type of Upper Layer PDU are same as IPv4's.

6. **Hop Limit** (8-bits): This field is used to stop packet to loop in the network infinitely. This is same as TTL in IPv4. The value of Hop Limit field is decremented by 1 as it passes a link (router/hop). When the field reaches infinitely.

7. **Source Address** (128-bits): This field indicates the address of originator of the packet.

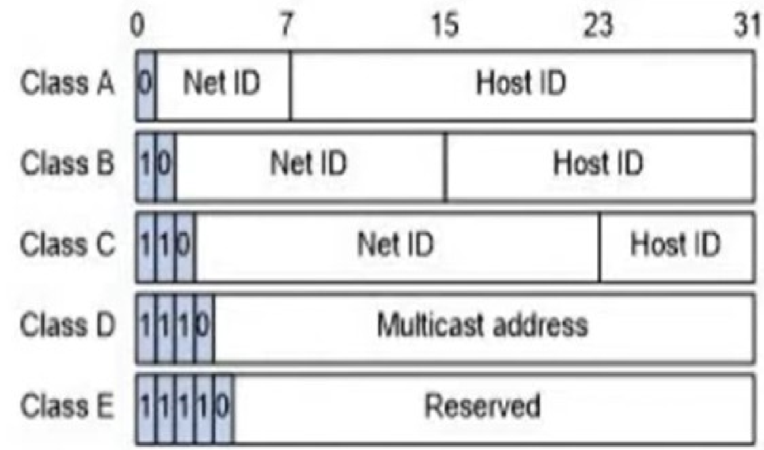
8. **Destination Address** (128-bits): This field provides the address of intended recipient of the packet.

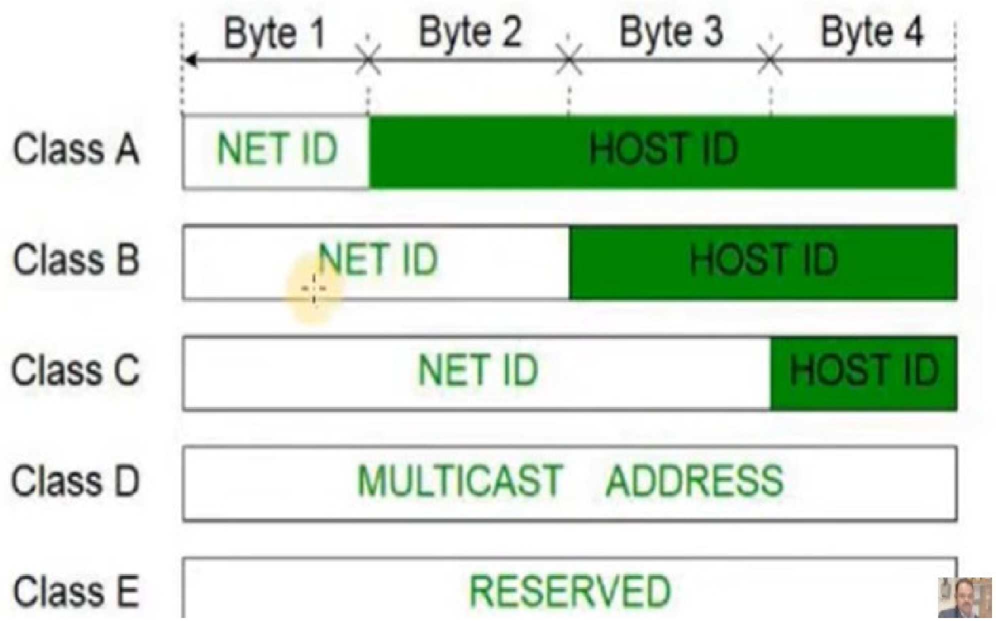
- **IP Addresses**
- TCP/IP version 4 or IPv4 uses 32-bit for logical address and IPv6 uses 128-bit for logical address.
- An IP address represented in dotted decimal notation. Example- 123.22.33.44
- IP address is divided into net id or network id and host id.
- IP Addresses are divided into five classes: Class A, Class B, Class c, Class C, Class D, Class E.

IP Address Class	Starting Binary Value	First Address	Last Address	No. of Network	No. of Host
Class A	0	1.0.0.0	126.255.255.254	2^7-1	2^{24} – 2
Class B	10	128.0.0.0	191.255.255.254	2^{14}	2^{16}-2
Class C	110	192.0.0.0	223.255.255.254	2^{21}	2^8-2
Class D	1110	224.0.0.0	239.255.255.254	Multicast	
Class E	1111	240.0.0.0	254.255.255.254	Undefined	

IP Address Classes

1. **Class A Address**
   a. Class id : 0
   b. Net ID : 7 bits
   c. Host id : 24 bits
2. **Class B Address**
   a. Class id : 10
   b. Net ID : 14 bits
   c. Host id : 16 bits
3. **Class C Address**
   a. Class id : 110
   b. Net ID : 21 bits
   c. Host id : 8 bits
4. **Class D addresses**
   a. Class id : 1110
   b. Multicast Addresses
5. **Class E addresses**
   a. Class id : 11110
   b. Reserved





Class **A** IP Addresses

- Class id : 0
- Sub-Networks
  - 2^{8-1} (128) networks
- Host in a Network
  - 2^{24} -2 (16 M) hosts in a Network
- First octet range : 1-126
- Start - End Address : 1.0.0.0 to 126.255.255.255
- 0 and 127 is reserved
  - **0.0.0.0**
    - Non routable IP address
    - Placeholder Address
    - Identify default route
  - **127.0.0.0 to 127.255.255.255**
    - Reserved loopback address
    - Localhost Address
    - Supported by Operating system
- Default subnet mask : /8 or 255.0.0.0

Class A

0NNNNNNN.HHHHHHHH.HHHHHHHH.HHHHHHHH

Class Id	0
Range	1-126
Networks	7 bits = 2^7=128
Hosts	24 bits = 2^{24}-2=16777214
Range	1.0.0.0 - 127.25.255.255
Subnet mask	/8 or 255.0.0.0
Example	100.10.12.206

Class B IP Addresses

- Class id : 10
- Sub-Networks
  - 2^{16-2} (126) networks
- Host in a Network
  - 2^{16} -2 (16384) hosts in a Network
- First octet range
  - 128-191
- Start - End Address
  - 128.0.0.0 to 191.255.255.255
- Default subnet mask
  - /16 or 255.255.0.0

		14 Bit	16 Bit
1	0	Network	Host

Class B

10NNNNNN.NNNNNNNN.HHHHHHHH.HHHHHHHH

Class Id	0
Range	128-191
Networks	14 bits = 2^{14}=16384
Hosts	16 bits = 2^{16}-2=65534
Range	128.0.0.0 - 191.255.255.255
Subnet mask	/16 or 255.255.0.0
Example	146.10.12.206

Class **C** IP Addresses

- Class id : 110
- Sub-Networks
  - 2^{24-3} =20,97,152 networks
- Host in a Network
  - 2^{8} -2=254 hosts in a Network
- First octet range
  - 192-223
- Start - End Address
  - 192.0.0.0 to 223.255.255.255
- Default subnet mask
  - /24 or 255.255.255.0

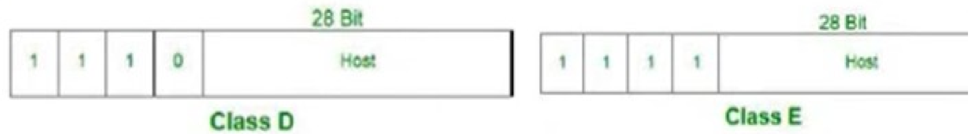
			21 Bit	8 Bit
1	1	0	Network	Host

Class C

110NNNNN.NNNNNNNN.NNNNNNNN..HHHHHHHH

Class Id	0
Range	192-223
Networks	21 bits = 2^{21}=20,97,152
Hosts	8 bits = 2^{8}-2=254
Range	192.0.0.0 - 223.255.255.255
Subnet mask	/24 or 255.255.255.0
Example	192.168.1.206

Class D & E IP Addresses



	Class D	Class E
Class Id	1110	1111
Range	224-239	240-255
Networks & Hosts	Not defined	Not defined
Range	224.0.0.0 - 239.255.255.255	240.0.0.0 - 255.255.255.255
Example	225.168.1.206	245.168.1.206
Use	Multicasting Addresses No Subnet Mask	Reserved Address R&D work Future use Not Subnet mask

IP Address Classes

Class	Class Id	First Octet Range	Network	Hosts	Format N-Network H-Host	Range
A	0	1-126	2^7-2=126	2^{24}-2=1,67,77,214	**N**.H.H.H	1.0.0.0 to 126.255.255.255
B	10	128-191	2^{14}=16,384	2^{16}-2=64,534	**N.N**.H.H	128..0.0 to 191.255.255.255
C	110	192-223	2^{21}=20,97,152	2^8-2=254	**N.N.N**.H	192.0.0.0 to 223.255.255.255
D	1110	224-239	Multicast Address			224.0.0.0 to 239.255.255.255
E	1111	240-255	R & D and Future use			240.0.0.0 to 255.255.255.255

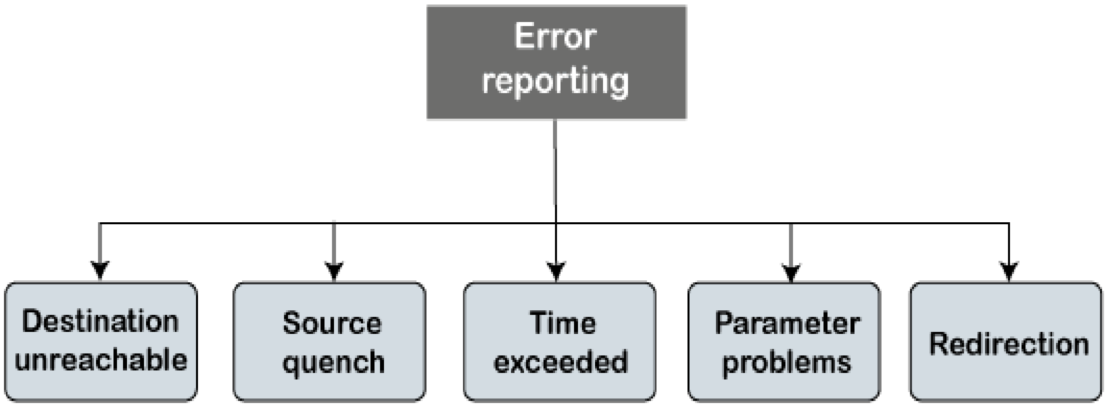
- **Internet Control Message Protocol (ICMP)**
- ICMP is network diagnostic and error reporting protocol.ICMP belongs to IP protocol suite and uses IP as carrier protocol. After constructing ICMP packet, it is encapsulated in IP packet. Because IP itself is a best-effort non-reliable protocol, so is ICMP.
- Any feedback about network is sent back to the originating host. If some error in the network occurs, it is reported by means of ICMP. ICMP contains dozens of diagnostic and error reporting messages.

Internet Control Message Protocol parameters

Message type	Description
Destination unreachable	Packet could not be delivered
Time exceeded	Time to live field hit 0
Parameter problem	Invalid header field
Source quench	Choke packet
Redirect	Teach a router about geography
Echo request	Ask a machine if it is alive
Echo reply	Yes, I am alive
Timestamp request	Same as Echo request, but with timestamp
Timestamp reply	Same as Echo reply, but with timestamp

- ICMP-echo and ICMP-echo-reply are the most commonly used ICMP messages to check the reachability of end-to-end hosts. When a host receives an ICMP-echo request, it is bound to send back an ICMP-echo-reply. If there is any problem in the transit network, the ICMP will report that problem.
- **ICMP Message Format**
- The message format has two things; one is a category that tells us which type of message it is. If the message is of error type, the error message contains the type and the code. The type defines the type of message while the code defines the subtype of the message

- **Types of Error Reporting messages**
- The error reporting messages are broadly classified into the following categories:



- **Destination unreachable**:The destination unreachable error occurs when the packet does not reach the destination. Suppose the sender sends the message, but the message does not reach the destination, then the intermediate router reports to the sender that the destination is unreachable.
- **Source quench**
- There is no flow control or congestion control mechanism in the network layer or the IP protocol. The sender is concerned with only sending the packets, and the sender does not think whether the receiver is ready to receive those packets or is there any congestion occurs in the network layer so that the sender can send a lesser number of packets, so there is no flow control or congestion control mechanism.

- **Time exceeded** :Sometimes the situation arises when there are many routers that exist between the sender and the receiver. When the sender sends the packet, then it moves in a routing loop. The time exceeded is based on the time-to-live value.
- When the packet traverses through the router, then each router decreases the value of TTL by one. Whenever a router decreases a datagram with a time-to-live value to zero, then the router discards a datagram and sends the time exceeded message to the original source.

- **Parameter problems:** The router and the destination host can send a parameter problem message. This message conveys that some parameters are not properly set.

- **Redirection :**When the packet is sent, then the routing table is gradually augmented and updated. The tool used to achieve this is the redirection message. For example, A wants to send the packet to B, and there are two routers exist between A and B. First, A sends the data to the router 1. The router 1 sends the IP packet to router 2 and redirection message to A so that A can update its routing table.

- Address Resolution Protocol, is a crucial network protocol that maps an IP address to a physical MAC address for devices on the same local network. It allows a device to find the MAC address of another device by sending an ARP request, which is then answered by the device with the matching IP address, enabling them to communicate at the data link layer. The device that made the request stores this IP-to-MAC mapping in its ARP cachefor future use, which improves network performance by reducing the need for repeated broadcasts.

- How ARP works
- **ARP Request**:
- When a device needs to send data to another device on the local network but only knows its IP address, it broadcasts an ARP request. This message asks, "Who has this IP address?".
- **ARP Reply**:
- Every device on the local network receives the request, but only the device with the matching IP address responds. It sends an ARP reply directly back to the requesting device with its own MAC address.

- **ARP Cache**:
- The requesting device receives the reply and stores the IP-to-MAC address pair in its local ARP cache. This is a temporary table that helps avoid sending another ARP request for the same IP address in the future.
- **Communication**:
- With the MAC address now known, the requesting device can properly format the data packet and send it to the correct destination on the local network.

Fig: ARP request is broadcast

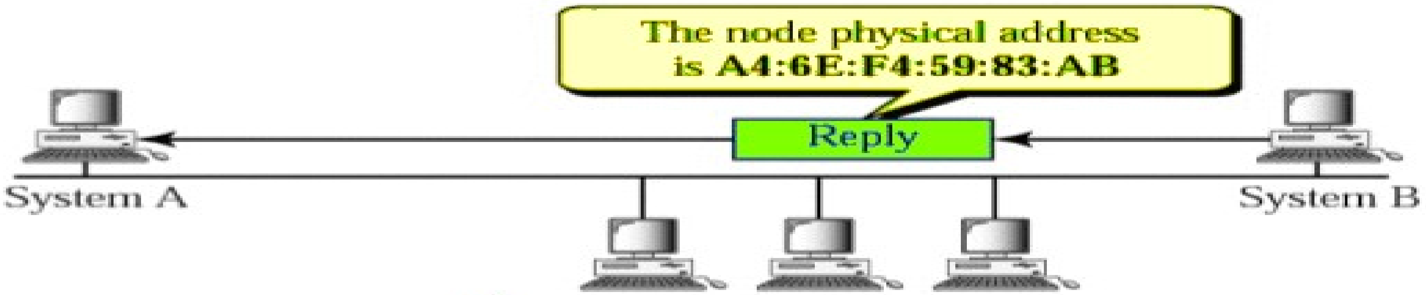
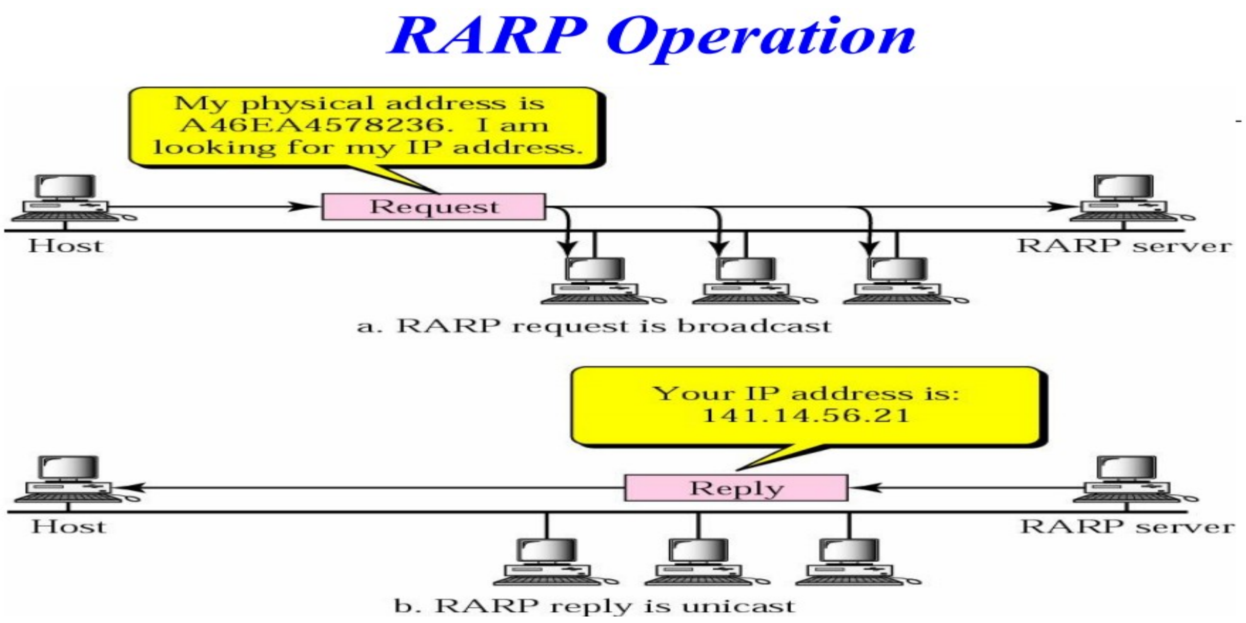


Fig: ARP reply is unicast

- Why ARP is important
- **Enables local communication**:
- ARP is fundamental for devices on the same network to find and communicate with each other, as it translates the logical IP address into the physical MAC address needed for data framing.
- **Improves efficiency**:
- By using an ARP cache, devices avoid continuously broadcasting ARP requests, which reduces network congestion and improves overall performance.
- **Supports flexibility**:
- It allows for dynamic mapping of addresses, meaning IP addresses can change (e.g., with DHCP) without requiring manual reconfigurations for each device's communication settings.

- Reverse Address Resolution Protocol, is a legacy network protocol that maps a device's physical MAC address to its IP address. It was primarily used by diskless workstations or other devices that lacked storage to find their own IP address on startup by broadcasting their MAC address and waiting for a RARP server to respond with the corresponding IP address. RARP has largely been replaced by more modern protocols like DHCP due to its limitations.
- How RARP works
- **Bootstrapping:** A device that doesn't know its IP address boots up and sends a RARP request packet with its MAC address to the network.
- **Broadcasting:** The request is broadcast to all devices on the local network.
- **Server lookup:** A dedicated RARP server that has a table mapping MAC addresses to IP addresses receives the request.
- **Response:** The RARP server looks up the requesting device's MAC address in its table and sends a RARP reply packet containing the corresponding IP address back to the device.

- Key differences from ARP
- **ARP (Address Resolution Protocol):**
- Maps an IP address to a MAC address (e.g., "What is the MAC address for IP address 192.168.1.1
- **RARP (Reverse ARP):**
- Maps a MAC address to an IP address (e.g., "What is the IP address for this MAC address?").

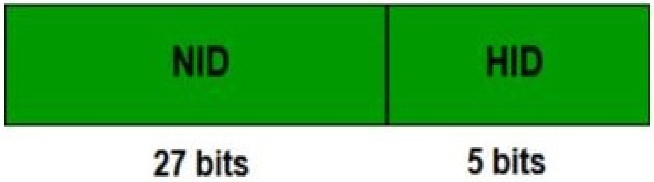


- **Classless Inter Domain Routing (CIDR)**
- CIDR(Classless Inter Domain Routing) is a method of IP address allocation and routing that allows more efficient use of IP addresses. Unlike traditional class based addressing, CIDR allocates IP addresses based on a network prefix rather than a fixed class (A, B, or C).
- **Notation:** a.b.c.d/n
- n = number of bits in the network prefix.
- Example: 192.168.1.0/24 first 24 bits are network, remaining 8 bits are host ID.

- **Why CIDR?**
- Classful addressing wastes IP addresses:
- Class IPs Available Hosts Example Wastage A 2^{24} =2^{24} - 2 Too large for small orgs B 2^{16} =2^{16} - 2 Wastes 49,150 hosts for 214 need
- **Rules for Forming CIDR Blocks**
- All IPs must be contiguous.
- Block size must be a power of 2 (2^n) simplifies network division.
- First IP of block divisible by block size least significant bits of host ID should be 0.
- **Example:** If the Block size is 2^5 then, Host Id will contain 5 bits and Network will contain 32 - 5 = 27 bits.
- edC$2^8 2^8$ - 2Small networks only

- First IP address of the Block must be evenly divisible by the size of the block. in simple words, the least significant part should always start with zeroes in Host Id. Since all the least significant bits of Host Id is zero, then we can use it as Block Id part.

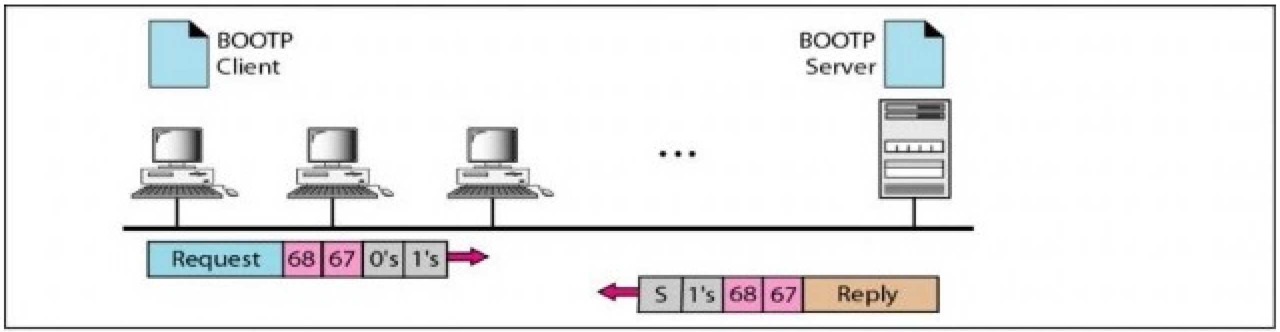
Example: If the Block size is 2^5 then, Host Id will contain 5 bits and Network will contain 32 - 5 = 27 bits.



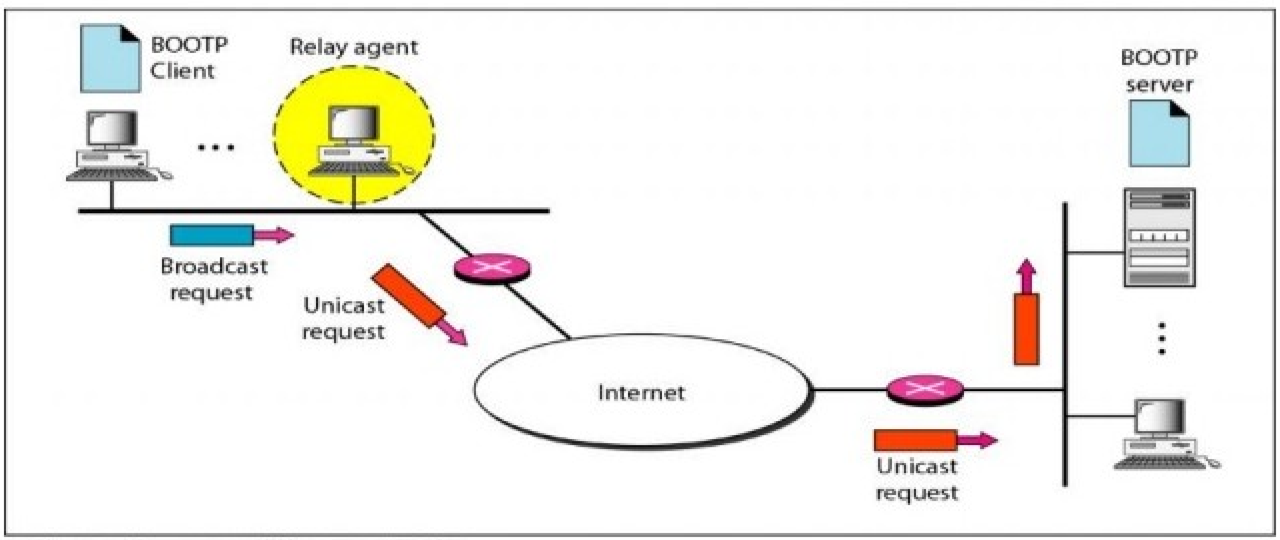
- **BOOTP**
- The Bootstrap Protocol (BOOTP) is a client/server protocol designed to provide physical address to logical address mapping. BOOTP is an application layer protocol. The administrator may put the client and the server on the same network or on different networks.
- One of the advantages of BOOTP over RARP is that the client and server are application-layer processes. As in other application-layer processes, a client can be in one network and the server in another, separated by several other networks. However, there is one problem that must be solved.

- The BOOTP request is broadcast because the client does not know the IP address of the server. A broadcast IP datagram cannot pass through any router. To solve the problem, there is a need for an intermediary. One of the hosts (or a router that can be configured to operate at the application layer) can be used as a relay. The host in this case is called a relay agent. The relay agent knows the unicast address of a BOOTP server.
- When it receives this type of packet, it encapsulates the message in a unicast datagram and sends the request to the BOOTP server. The packet, carrying a unicast destination address, is routed by any router and reaches the BOOTP server. The BOOTP server knows the message comes from a relay agent because one of the fields in the request message defines the IP address of the relay agent. The relay agent, after receiving the reply, sends it to the BOOTP client.

- BOOTP

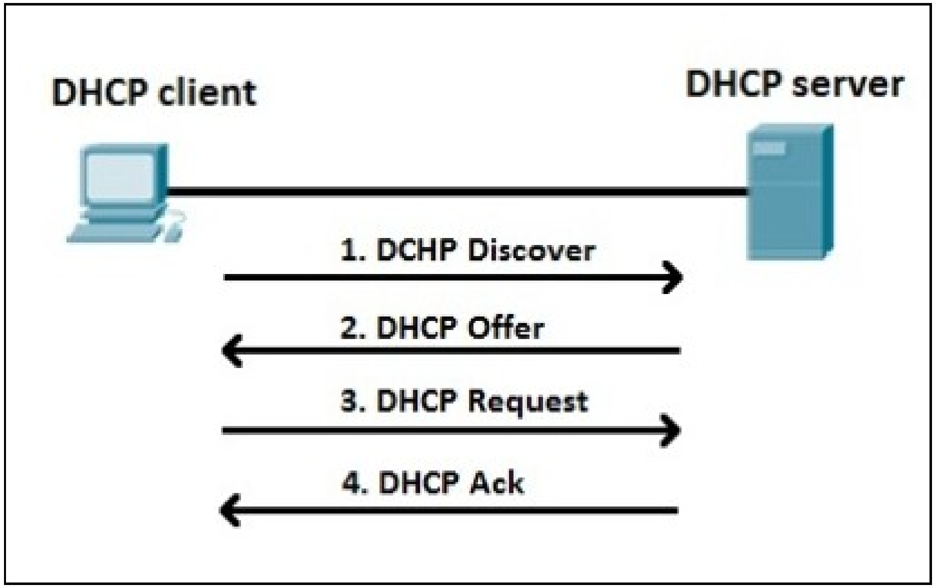


a. Client and server on the same network



b. Client and server on different networks

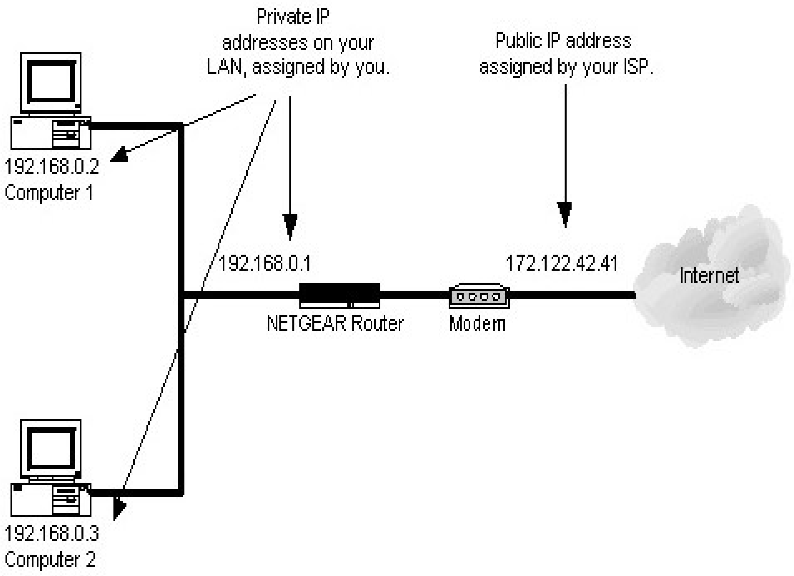
- DHCP :Dynamic Host Configuration Protocol is a network protocol that automates the assignment of IP addresses and other network configuration details to devices on a network. Operating on a client-server model, a DHCP server assigns these settings from a pool of available addresses to DHCP clients. This process eliminates the need for manual configuration, reduces errors like IP address conflicts, and streamlines network management for both IPv4 and IPv6 networks.



- How DHCP Works (The DORA Process)
- DHCP uses a four-phase process often abbreviated as DORA:
- **DHCP Discover:** A new device (client) that joins a network broadcasts a "DHCP Discover" message to find a DHCP server.
- **DHCP Offer:** A DHCP server that receives the discover message responds with a "DHCP Offer," proposing an IP address and other configuration details.
- **DHCP Request:** The client accepts the offer by broadcasting a "DHCP Request" message, indicating which server's offer it is accepting.
- **DHCP Acknowledge:** The server sends a "DHCP Acknowledge" (ACK) message, confirming the IP address assignment and providing the client with its lease information.

- Network Address Translation (NAT) is a networking technique that allows multiple devices within a private network to access external networks (like the Internet) using a single public IP address. By translating private IP addresses into public IP addresses and vice versa, NAT conserves the limited pool of IPv4 addresses and adds a layer of security by masking internal addresses from the outside world.

Given below is the diagram of the NAT



- Understanding Network Address Translation (NAT)
- Now let us try to understand what Network Address Translation (NAT) is:
- **Step 1** Consider you have internet provided by Internet Service Provider ABC.
- **Step 2** So, they will give you connection to your Modem. That connection we used to call WAN.
- **Step 3** This connection is always configured with a Public IP address.
- **Step 4** Then, your LAN side of the MODEM is configured with a Private IP address.
- **Step 5** That means your computer or laptop connected to the network receives a Private IP address.
- **Step 6** As per the standard Private IP will not communicate with Public IP address at any Point of time.
- **Step 7** To achieve this, Private IP addresses need to be translated to Public IP addresses with help of NAT.
- **Step 8** In simple words, Network Address translation is used to translate Private IP address to Public IP address to communicate LAN side of the Device to Global Network. Network address translation can be processed in Router or Firewall.

- Working of NAT
- Usually we used gateway router / Border devices used for NAT configuration. One of the interfaces for that device is connected to the local Area network (INSIDE) and one of the interfaces for this device connected to the outside network (OUTSIDE).
- When we have received a request from our local machine it will hit the configuration pool then that Private IP will convert it into Public IP address and vice versa.
- **Inside worldwide location** IP address that speaks to at least one inside nearby IP delivers to the rest of the world. This is within have as observed from the external organization.
- **Outside residential area** This is the genuine IP address of the objective host in the nearby organization after interpretation.
- **Outside worldwide location** This is the external host as observed to structure the external organization. It is the IP address of the external objective host before interpretation.